**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**
**Department of Computer Science and Engineering**
**CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop**

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| **Course code:** | CSA10 / CSA1063 | **Course title:** | Software Engineering |
| **CO Assessed:** | CO1 | **Assessment:** | Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop** |
| **Weightage:** | 20% | **Total marks:** | 25 |
| **CO1 Statement:** | *Apply advanced methodologies and agile project management techniques to manage software projects effectively* | | |
| **Student name:** | Amirtha Sri N | **Reg No:** | 1911260175 |
| **Date:** | 9.10.2026 | **Duration:** | 60 minutes |

**Code of Conduct**

*I Amirtha sri N (1911260175)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the Student: _____

**Annexure A – Agile Sprint Planning & User Story Workshop**
**Instructions to Students:**
Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:
Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format:
     *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint

**Rubrics for Calculation**

| Criteria | Excellent (5 Marks) | Good (3 Marks) | Satisfactory (2 Marks) | Needs Improvement (1 Mark) |
|---|---|---|---|---|
| **Project Vision & Stakeholder Identification (30%)** | Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified. | Vision is clear with minor omissions. Most stakeholders are correctly identified. | Vision is partially defined. Some stakeholders are missing or incorrectly identified. | Vision is unclear or incomplete. Stakeholders are poorly identified or missing. |
| **Epics & User Stories (25%)** | Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements. | Epics and user stories are mostly complete with minor formatting or requirement issues. | User stories are partially complete with limited Agile structure. | User stories are incomplete, unclear, or do not follow Agile practices. |
| **Acceptance Criteria (20%)** | Acceptance criteria are specific, measurable, testable, and aligned with all user stories. | Acceptance criteria are mostly clear and testable with minor gaps. | Acceptance criteria are partially defined and lack clarity or completeness. | Acceptance criteria are missing or not aligned with user stories. |
| **Product Backlog Prioritization & Sprint Planning (15%)** | Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete. | Backlog prioritization and sprint planning are mostly appropriate with minor issues. | Prioritization or sprint planning is partially complete with limited justification. | Backlog prioritization and sprint planning are poorly organized or incomplete. |
| **Story Point Estimation, Sprint Goal & Presentation (10%)** | Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized. | Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear. | Estimation or sprint goal lacks justification. Presentation is adequate but incomplete. | Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete. |

| Criteria | **Mark** |
|---|---|
| Project Vision & Stakeholder Identification **(30%)** | /5 |
| Epics & User Stories **(25%)** | /5 |
| Acceptance Criteria **(20%)** | /5 |
| Product Backlog Prioritization & Sprint Planning **(15%)** | /5 |
| Story Point Estimation, Sprint Goal & Presentation **(10%)** | /5 |
| **TOTAL** | /25 |

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# 1. Project Vision and Objectives

**Project Vision:** The vision of the AI-Based College FAQ and Student Support Chatbot is to provide students with quick, reliable, and easy access to common college information through a conversational digital assistant. The chatbot will answer frequently asked questions about admissions, departments, class schedules, examinations, fees, attendance, events, and campus services. It will reduce repetitive enquiries handled by college staff and help students find information at any time.

**Problem Statement:** Students often need to contact different departments or search multiple notices and websites to find basic college information. Replies may be delayed, especially outside office hours. A centralized chatbot can make frequently requested information easier to access while directing complicated or personal enquiries to the appropriate staff member.

## Objectives

• Provide answers to common college-related questions in simple language.
• Make information available 24/7 for supported FAQs.
• Reduce repetitive questions received by administrative staff.
• Provide a clear option to contact or escalate an unanswered query to the relevant department.
• Keep FAQ content manageable so authorized staff can update it.
• Protect student privacy and avoid displaying confidential personal records.

**Project Scope:** The first version will support text-based questions and answers from an approved FAQ knowledge base. It will include topic categories, basic question matching, fallback responses, and an escalation/contact option. Student-specific functions such as viewing individual marks, fee balances, or personal attendance are outside the initial scope unless secure college-system integration is approved.

## Key Stakeholders

| Stakeholder | Role / Interest |
|---|---|
| Students | Main users; ask questions and evaluate whether answers are useful. |
| College administration | Provides and approves official information about rules, fees, schedules, and services. |
| Department staff / faculty | Provide department-specific answers and handle escalated questions. |
| Project team / developers | Design, develop, test, and maintain the chatbot. |
| Project guide / product owner | Clarifies requirements, prioritizes features, and accepts completed work. |
| IT administrator | Supports hosting, access control, security, and maintenance. |

**Success Measures:** The prototype should correctly answer the approved test FAQs, show a useful fallback when it cannot answer, allow users to find contact details, and pass basic usability and privacy checks.

# 2. Epics and User Stories

An epic is a broad area of functionality. Each epic is divided into smaller user stories that can be developed and tested during a sprint.

| Epic | User Story ID | User Story |
|---|---|---|
| Epic 1: Ask a Question | US1 | As a student, I want to type a question in a chat window, so that I can quickly request college information. |
| Epic 1: Ask a Question | US2 | As a student, I want to see a clear answer to a supported FAQ, so that I can understand the information without searching several sources. |

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|---|---|---|
| Epic 2: Find FAQ Topics | US3 | As a student, I want to browse FAQ categories, so that I can find relevant information when I do not know the exact wording. |
| Epic 3: Handle Unknown Questions | US4 | As a student, I want the chatbot to tell me when it cannot answer and show a contact option, so that I know what to do next. |
| Epic 4: Maintain FAQ Content | US5 | As an authorized staff member, I want to add or update approved FAQs, so that chatbot responses remain current. |
| Epic 5: Privacy and Safety | US6 | As a student, I want the chatbot to avoid showing private student records, so that my personal information remains protected. |

# 3. Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied for a user story to be considered complete. The following criteria use the Given–When–Then format.

### US1 – Type a question

**Given** the chatbot page is open, **When** the student enters a non-empty question and selects Send, **Then** the question should appear in the chat and the system should process it. Empty input should not be submitted.

### US2 – Receive an FAQ answer

**Given** the student asks a question that matches an approved FAQ, **When** the chatbot processes the question, **Then** it should display the corresponding approved answer in the chat.

### US3 – Browse categories

**Given** the student opens the FAQ categories, **When** the student selects a category such as Admissions, Examinations, or Attendance, **Then** the system should display relevant questions for that category.

### US4 – Unknown question fallback

**Given** the question does not match the available FAQ content, **When** the chatbot evaluates it, **Then** it should clearly state that it cannot find a reliable answer and display a college contact or escalation option.

### US5 – Update FAQ content

**Given** an authorized staff member has access to the FAQ management function, **When** the staff member saves a valid FAQ update, **Then** the new or edited FAQ should be stored and available to the chatbot. Unauthorized users must not be able to edit FAQs.

### US6 – Protect privacy

**Given** a user asks for private information such as another student's marks or personal details, **When** the chatbot processes the request, **Then** it must not disclose that information and should direct the user to an authorized college channel.

# 4. Product Backlog and Prioritization

The product backlog is ordered by user value, risk, and dependency. Priority labels are High, Medium, and Low. The first sprint focuses on a usable FAQ chatbot prototype.

| Priority | Story | Feature / Requirement | Reason |
|---|---|---|---|
| 1 – High | US1 | Chat input and send question | Required before users can ask questions. |
| 2 – High | US2 | Match common questions and display answers | Core purpose of the chatbot. |

| | | | |
|---|---|---|---|
| 3 – High | US4 | Fallback response and contact option | Prevents dead ends when no answer is available. |
| 4 – Medium | US3 | Browse FAQ categories | Improves discoverability; depends on FAQ content being organized. |
| 5 – Medium | US6 | Privacy protection rules | Essential safeguard to test from the prototype stage. |
| 6 – Medium | US5 | Authorized FAQ editing | Helps staff maintain content; can be developed after the basic chat flow. |

# 5. Sprint Backlog – Sprint 1

**Sprint Duration:** 2 weeks (proposed).

**Sprint Goal:** Build a working chatbot prototype that accepts a student's question, answers supported college FAQs, and gives a helpful fallback/contact option when it cannot answer.

| Selected Story | Development Tasks | Expected Output |
|---|---|---|
| US1 – Chat input | Create chat screen; add text input and Send button; display student message; validate empty input. | Student can submit a question. |
| US2 – FAQ answers | Prepare a small approved FAQ dataset; implement question matching; return the mapped answer; test sample questions. | Supported questions receive relevant answers. |
| US4 – Fallback | Define fallback message; add department contact/escalation information; test unmatched questions. | Unknown questions receive next-step guidance. |
| US6 – Privacy basics | Avoid retrieving private student records; add safe response for requests for confidential information; test privacy-related prompts. | Prototype does not expose private student data. |

## Sprint Workflow

1. Sprint Planning: confirm the sprint goal, choose stories, and agree on acceptance criteria.
2. Design: sketch the chat interface and define the FAQ data format.
3. Development: implement input, FAQ matching, fallback, and basic privacy rules.
4. Testing: test supported questions, empty input, unmatched questions, and privacy-related questions.
5. Sprint Review: demonstrate the working prototype to the project guide or stakeholders.
6. Retrospective: discuss what went well, what was difficult, and what to improve in Sprint 2.

# 6. Story Point Estimation

Story points estimate relative effort, complexity, and uncertainty rather than exact hours. A Fibonacci-style scale (1, 2, 3, 5, 8) is used. These are proposed estimates for a student prototype and may be adjusted after team discussion.

| User Story | Points | Justification |
|---|---|---|
| US1 – Chat input and send | 3 | Moderate effort: interface, input validation, and displaying messages. |
| US2 – FAQ answer matching | 5 | Higher effort: preparing FAQ data, matching questions, and testing answer relevance. |
| US4 – Fallback and contact option | 2 | Small feature with a clear response and contact details. |

| | | |
|---|---|---|
| US6 – Basic privacy rules | 3 | Requires identifying sensitive requests and verifying that private information is not returned. |
| Total Sprint 1 | 13 | A manageable proposed workload for a small student project team. |

### Estimation Technique

Planning Poker can be used by team members. Each member selects a point value for a story, the team discusses differences, and a shared estimate is agreed. For example, US2 receives 5 points because FAQ matching and answer testing are more complex than adding a basic contact message.

## 7. Sprint Goal and Expected Deliverables

**Sprint Goal:** Deliver a basic, demonstrable chatbot that lets students submit college-related questions, returns answers for a selected set of approved FAQs, provides a fallback for unknown questions, and follows basic privacy safeguards.

### Expected Deliverables at the end of Sprint 1

- A simple chat interface with a question input field and Send button.
- A starter FAQ dataset covering common topics such as admissions, attendance, examinations, departments, andcampus services.
- Question-to-answer matching for the selected FAQs.
- A fallback response with a relevant college contact or escalation route.
- Basic privacy handling for requests involving confidential student information.
- Test cases and results for supported questions, empty input, unknown questions, and privacy-related prompts.
- A short demonstration of the working prototype.

### Definition of Done

A story is considered done when its code or configuration is completed, acceptance criteria are met, relevant tests pass, the feature is demonstrated, and any known limitation is documented. FAQ content must be checked against approved college information before being used.

## 8. Risks and Future Improvements

**Risks:** Outdated FAQ answers, ambiguous student questions, incorrect answer matching, and attempts to request confidential information. These risks can be reduced by using approved content, testing common question variations, providing a fallback instead of guessing, and enforcing access controls.

**Future Improvements:** Add multilingual support, improve natural-language understanding, provide a staff dashboard, include feedback ratings, expand FAQ categories, and integrate with official college systems only after security, privacy, and authorization requirements are reviewed.

## Conclusion

Agile sprint planning breaks the chatbot project into small, testable features and allows the team to deliver useful functionality incrementally. The prioritized backlog, user stories, acceptance criteria, story-point estimates, and sprint goal provide a clear plan for building a reliable and student-friendly college FAQ and support chatbot.

**Reference:** CSA10 / CSA1063 Software Engineering, CO1 Assessment Tool 2 – Agile Sprint Planning & User Story Workshop. Project-specific content prepared for the topic “AI-Based College FAQ and Student Support Chatbot.”